

CARDBOARD BOAT

BUILD PLANS

Step-by-Step Instructions for a Lightweight, Race-Ready Boat

WHAT'S INCLUDED

- ✓ Multi-Step Assembly Sheets
- ✓ Illustrated Diagrams
- ✓ Material & Build Guidance
- ✓ Beginner-Friendly Design

Lightweight. Strong. Proven Race-Winning Design.

BEFORE YOU START

✓ Read through all steps before cutting.

- ✓ Do a full dry fit before taping (critical)
- ✓ Use double-wall cardboard for best results
- ✓ Build on a flat surface



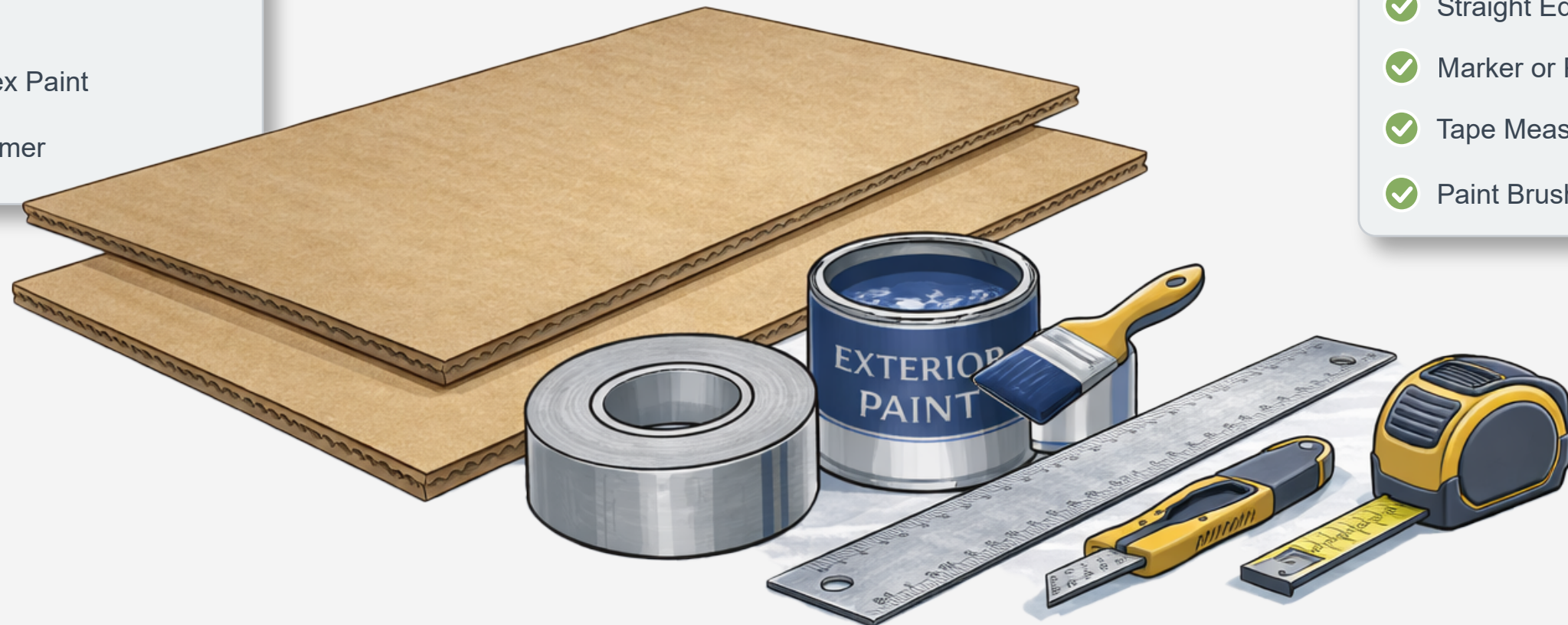
Tip: This build is designed to be **simple** — but alignment matters. Take your time on the first steps.

MATERIALS & TOOLS

- ✓ Gather all materials before starting your build. Having everything ready will make assembly faster and easier.

MATERIALS

- ✓ Cardboard Sheets
(double-wall recommended)
- ✓ Duct Tape
- ✓ Exterior Latex Paint
- ✓ Optional: Primer



TOOLS

- ✓ Utility Knife / Box Cutter
- ✓ Straight Edge / Ruler
- ✓ Marker or Pencil
- ✓ Tape Measure
- ✓ Paint Brush / Roller

MATERIAL NOTES

- ✓ Use double-wall cardboard for strength
- ✓ Avoid damaged or wet cardboard
- ✓ Larger sheets reduce seams and improve strength

TOOL TIPS

- ✓ Use sharp blades for clean cuts
- ✓ Replace blades as needed
- ✓ Use straight edge for accuracy
- ✓ Measure twice & cut once

AVOID THESE

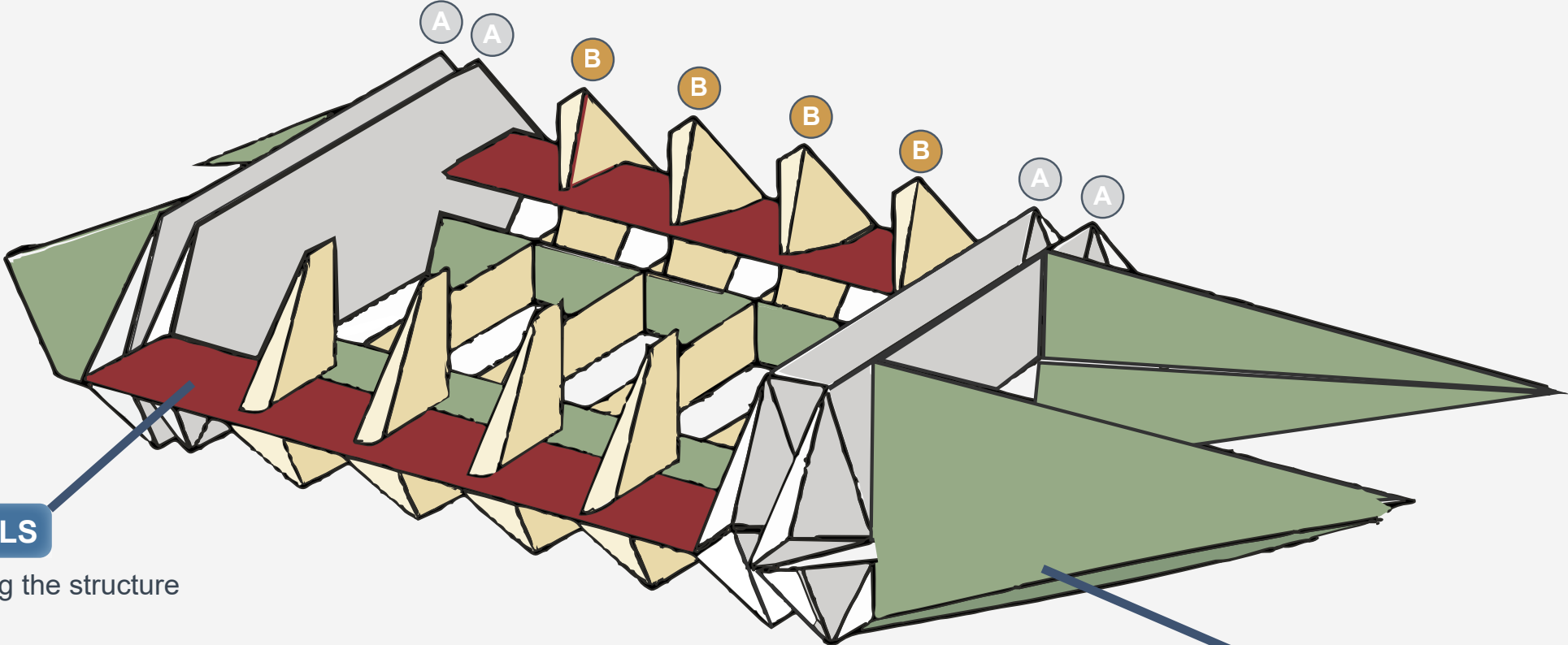
- ✗ Using thin cardboard
- ✗ Starting without all materials
- ✗ Using dull blades (causes rough cuts)
- ✗ Skipping straight edge cuts

STEP 1 | BUILD INTERNAL SKELETON

✓ This internal structure sets the shape, strength, and alignment of your boat.
Do a full dry fit before tape — alignment is critical.

PARTS REQUIRED

- Center Spine Panels (2)
- Side Support Rails (2)
- Internal Panels A (4)
- Internal Panels B (4)
- Cockpit Side Panels (2)
(placement reference only)



SIDE SUPPORT RAILS

Lock into each slot along the structure

INTERNAL PANELS

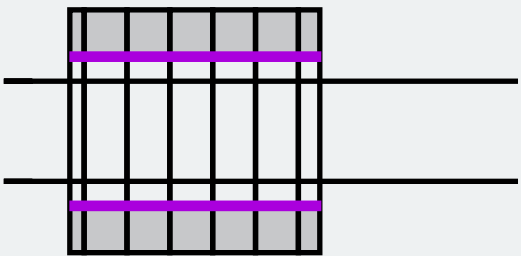
Insert in this order



CENTER SPINE PANELS

Make sure they are parallel

✓ COCKPIT PANELS PLACEMENT



- Sitting flush on internal supports (shown in purple)

✓ DRY-FIT CHECKLIST

- ✓ All panels fully seated in slots
- ✓ All panels level and parallel
- ✓ Structure is symmetrical (left/right)
- ✓ "A" panels at both ends, "B" panels in center

! DO NOT TAPE YET

- ✗ Key panels must be aligned perfectly first
- ✗ Fix spacing issues before taping
- ✗ Ensure panels sit square before taping

✓ **Final Tip:** Work from the center outward and recheck alignment after each panel is added.

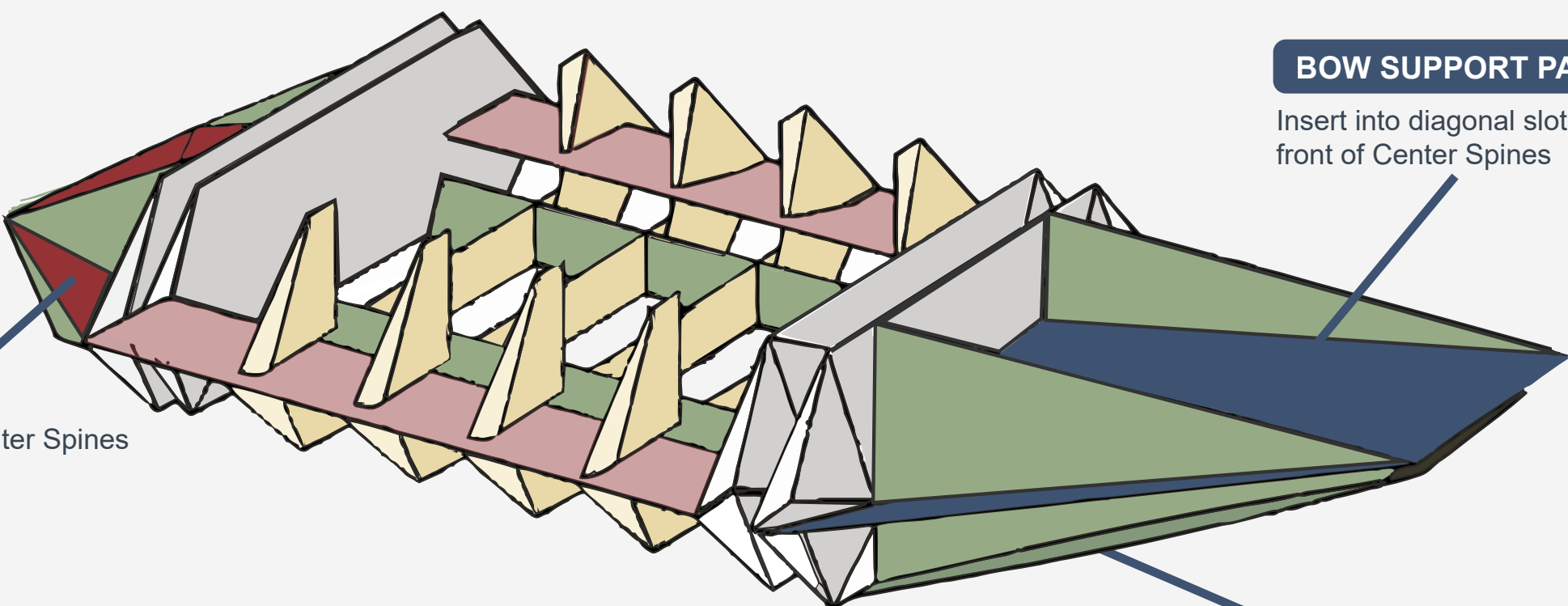
STEP 2 | ADD THE BOW AND STERN SUPPORTS



This is where we complete the internal construction by adding bow and stern supports.
Install fully before securing - Alignment is critical.

PARTS REQUIRED

- Bow Support Panel
- Stern Support Panel



STERN SUPPORT PANEL

Insert into slots at the back of Center Spines

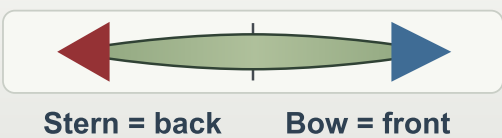
BOW SUPPORT PANEL

Insert into diagonal slots at the front of Center Spines

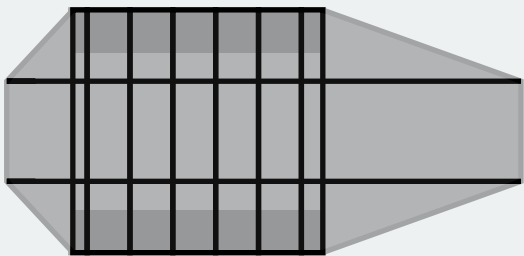
CENTER SPINE PANELS

Keep them parallel

BOW vs STERN



TOP VIEW VERIFICATION



• Boat should look like this from Top View



SUPPORT INSTALL CHECKLIST

- All panels fully seated in slots
- All panels are centered
- Panels aligned with center spines
- No gaps at slot connections



DO NOT SECURE YET

- Do not tape at this stage
- Ensure both panels are fully seated
- Final securing occurs in the next stage



Final Tip: Insert panels carefully - adjust alignment if resistance occurs. Do not force.

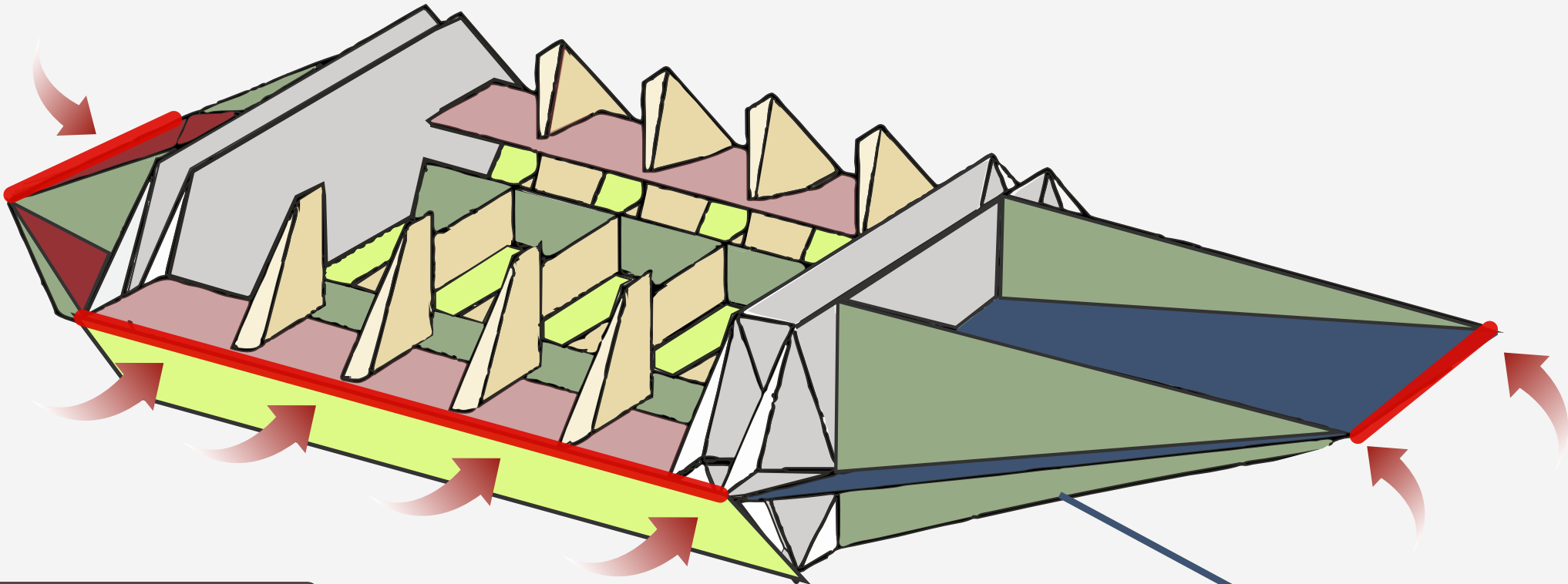
STEP 3 | ATTACH BOTTOM PANEL



This step locks the entire structure into final position.
Ensure full contact before securing - alignment is critical.

PARTS REQUIRED

- Bottom Panel (1)
- Tape



TAPE ALONG SEAMS

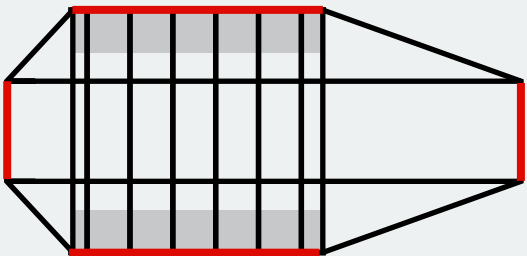
Apply tape along all seams. Start at center and work outward

BOTTOM PANEL

Align with internal skeleton and seat onto all supports



TOP VIEW VERIFICATION



• Taping shown here in red.



INSTALL CHECKLIST

- Panel fully seated on all ribs
- No gaps between panel and structure
- Edges aligned with frame
- Structure remains square and level



ALIGN BEFORE FULL SECURE

- Do not tape until alignment is confirmed
- Correct gaps or misalignment first
- Final securing happens AFTER verification



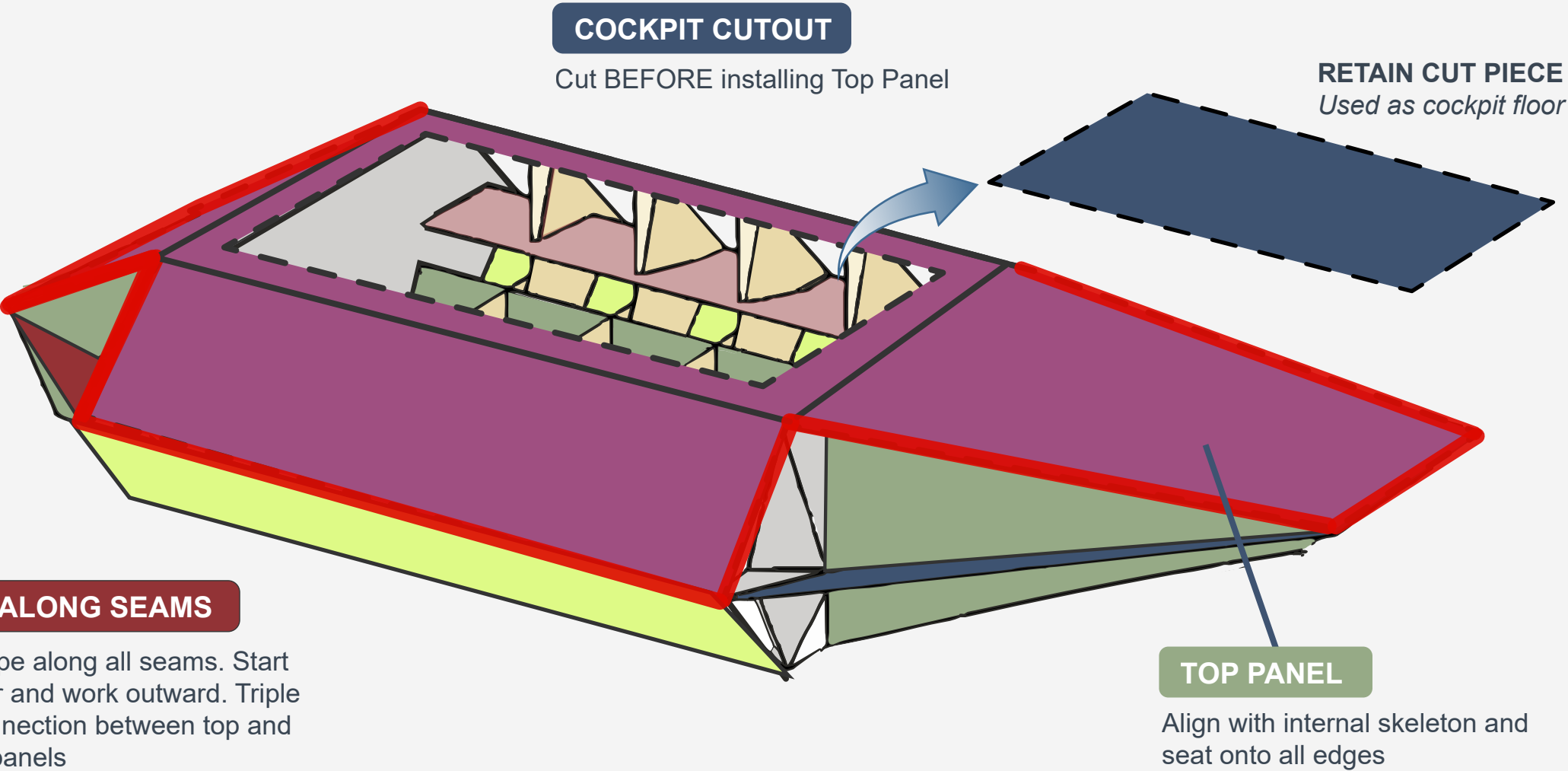
Final Tip: Apply even pressure and confirm alignment before fully taping.

STEP 4 | ATTACH TOP PANEL

✓ **Cut cockpit opening before installing top panel.**
Keep cutout piece - it becomes the cockpit floor.

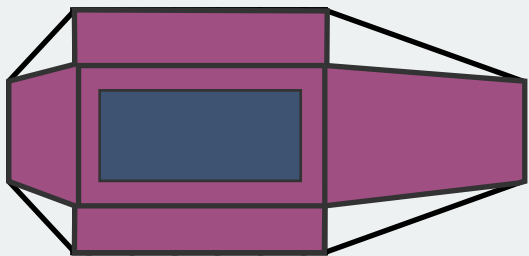
● **PARTS REQUIRED**

- Top Panel (1)
- Tape
- Cockpit Cutout (keep)



TAPE ALONG SEAMS
Apply tape along all seams. Start at center and work outward. Triple tape connection between top and bottom panels

✓ **TOP VIEW VERIFICATION**



• Top panel shown in purple. Cut out in blue.

✓ **INSTALL CHECKLIST**

- ✓ Top panel fully seated along all edges
- ✓ Cockpit opening clean and centered
- ✓ Top panel aligns with bottom panel edge
- ✓ No gaps along seams.

! **COCKPIT - CUT OUT FIRST**

- ✗ Do not install top panel until cutout removed.
- ✗ Do not discard cutout piece.
- ✗ Place cutout as cockpit floor.

✓ **Final Tip:** Apply even pressure and confirm alignment before fully taping.

STEP 5 | CREATE & ATTACH SIDE PANELS

✓ Overlay cardboard sheet and trace the opening. Then cut to shape (easier than measuring angles).

PARTS REQUIRED

- Side Panels (2)
(mirrored)
- Tape

TRACE METHOD (recommended)

Trace opening directly from boat. Cut and trim as needed.

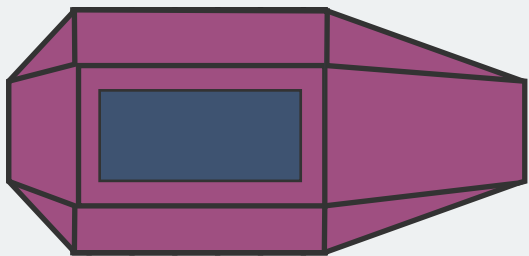
SIDE PANELS

Trace opening, cut to fit and use panel as template. Trim as needed. Taped edges will hide small gaps.

TAPE ALONG SEAMS

Apply tape along all seams.

✓ TOP VIEW VERIFICATION



- Side panels align cleanly with top & bottom edges

✓ INSTALL CHECKLIST

- ✓ Use the first panel as a template
- ✓ No large gaps (small gaps OK)
- ✓ Side panels sit flush on all sides
- ✓ Bow and stern transitions are smooth

! TEST FIT BEFORE TAPING

- ✗ Do not tape before confirming fit.
- ✗ Trim panels as needed
- ✗ Taped edges will hide minor imperfections

✓ **Final Tip:** Trace once, cut carefully and test fit before taping. Use the first panel as a template for the second.

STEP 6 | WATERPROOF & REINFORCE

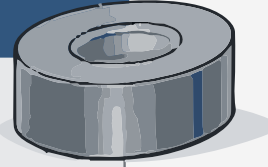
✓ Seal all seams with tape and apply paint - do not over-tape.

● PARTS REQUIRED

- Exterior Paint
- Tape
- Optional: Primer

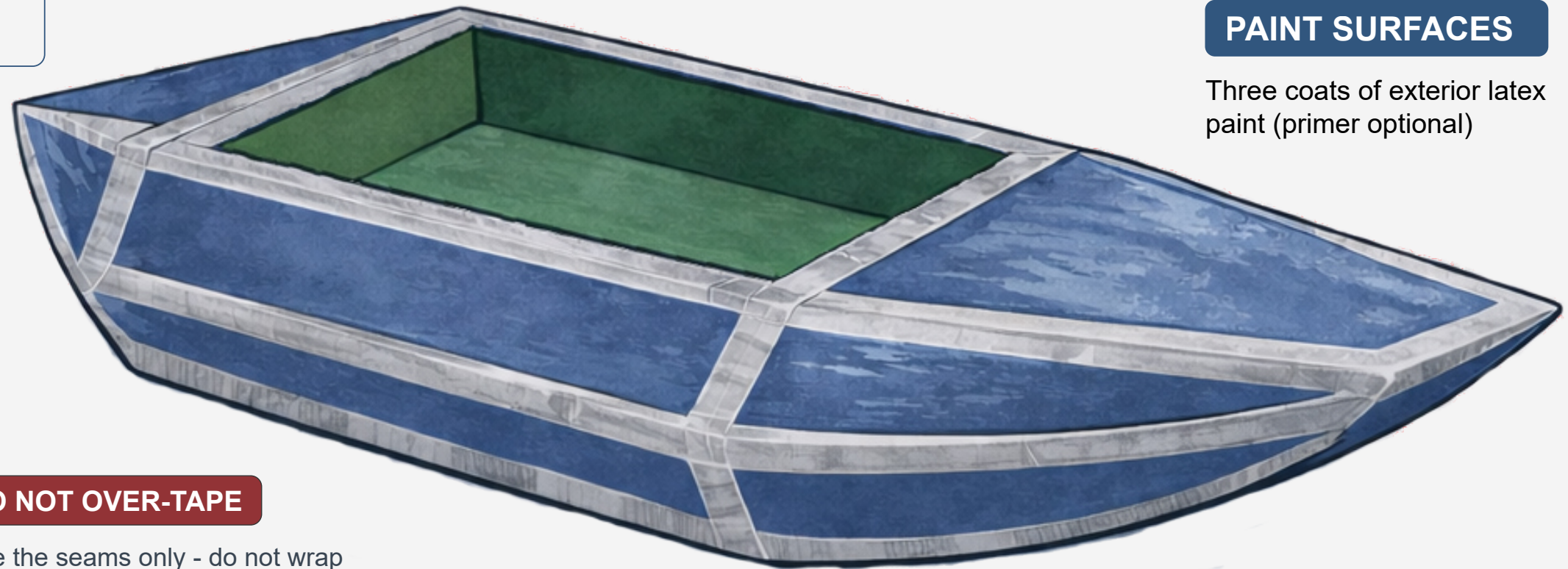
REINFORCE SEAMS

Seal and reinforce all seams with tape



PAINT SURFACES

Three coats of exterior latex paint (primer optional)



DO NOT OVER-TAPE

Tape the seams only - do not wrap the entire boat with tape.

✓ FINAL CHECKLIST

- ✓ All seams are sealed
- ✓ No exposed cardboard edges
- ✓ Cockpit edges sealed

✓ PAINT

- ✓ Apply evenly. Optional: start with primer
- ✓ Minimum of three coats of paint
- ✓ Use exterior latex paint
- ✓ Let dry between coats



! DO NOT OVER-TAPE

- ✗ Tape the seams only
- ✗ Do not wrap entire boat

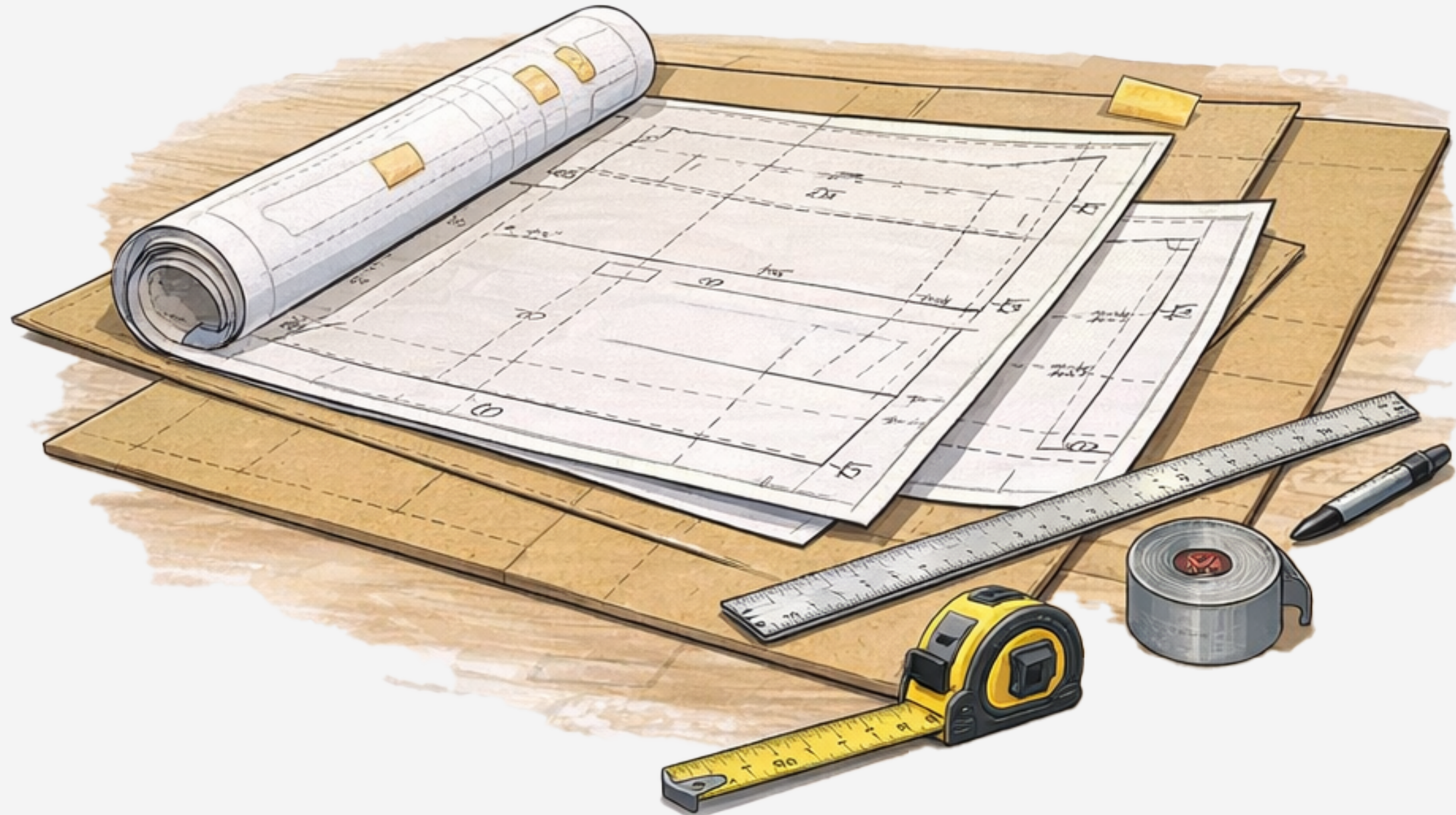


✓ **Final Tip:** Once seams are sealed, apply a primer coat (optional) and three coats of exterior latex paint.

PANEL TEMPLATES & CUT SHEETS

The following pages include exact panel templates used in this build.

Cut directly using these measurements

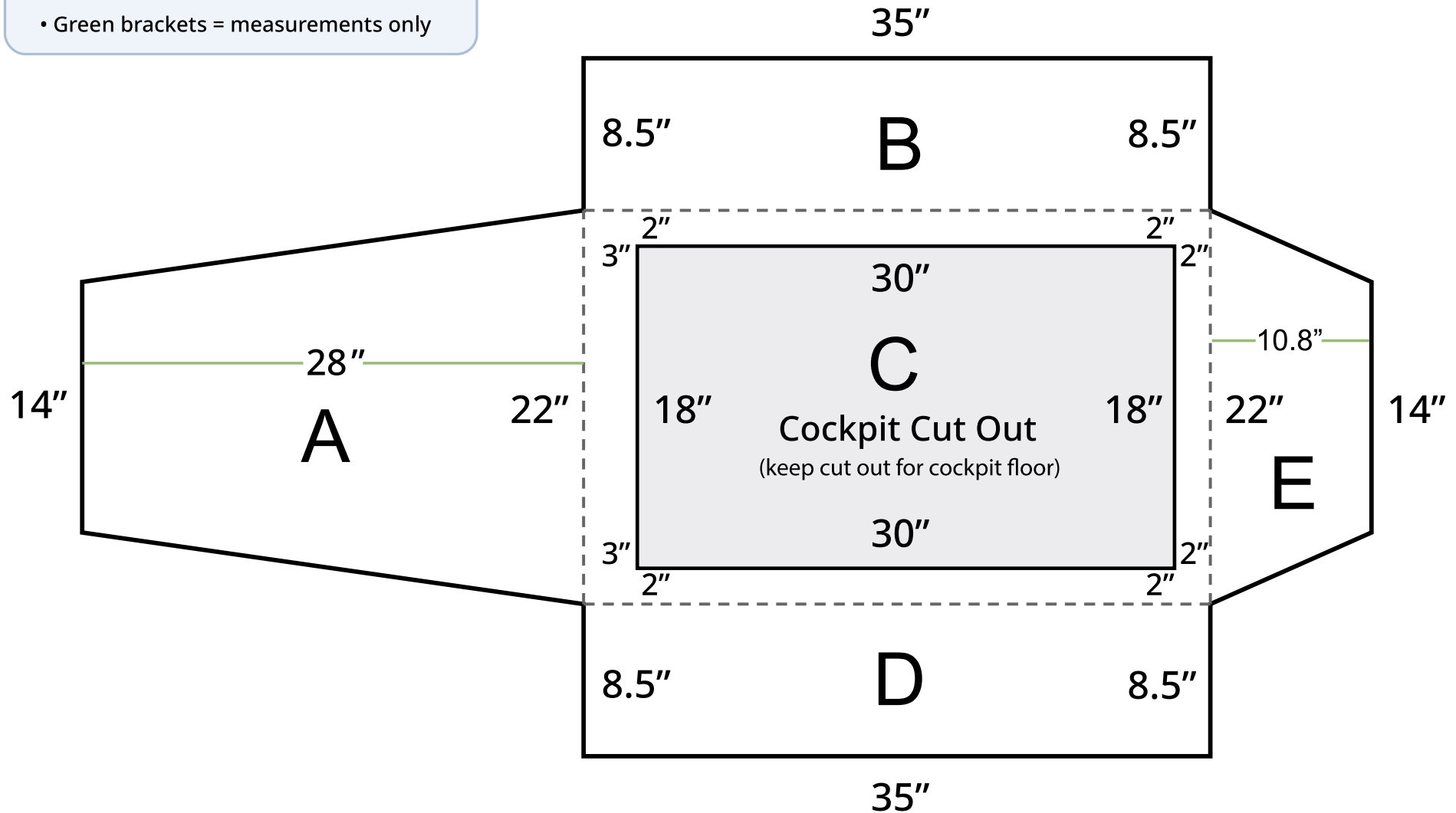


Measure twice and cut once

Top Panel Cut Template

Key

- Solid black = cut lines
- Dashed gray = fold / score lines
- Green brackets = measurements only

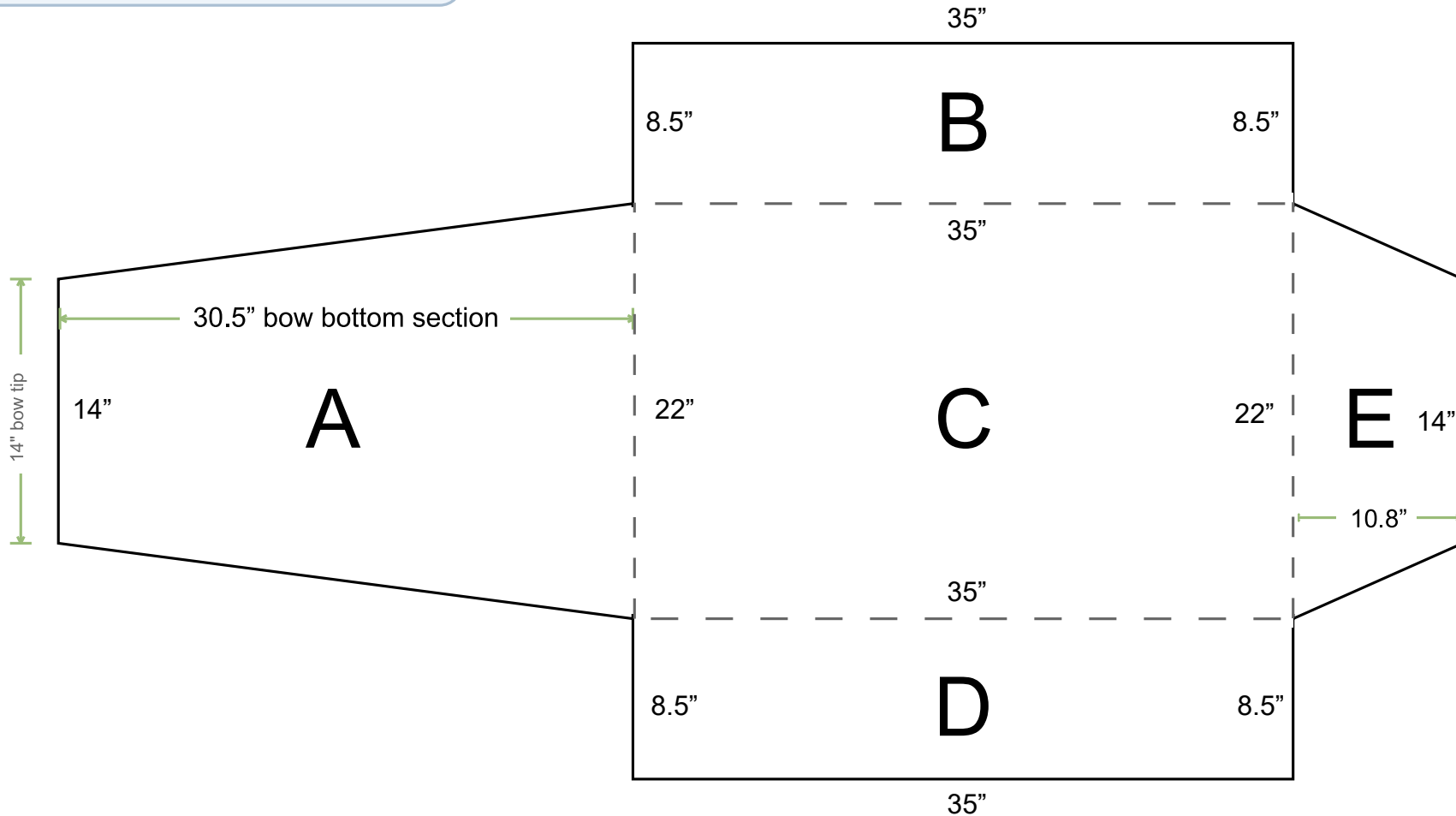


Pro Tip: This design is strongest when this panel is cut from one sheet (scored). But if that isn't available, you can cut these from smaller pieces (labeled A-E here and duct tape together) **Panel C Notes** (Full Panel is 22"x35, Cockpit is 18"x30")

Bottom Panel Cut Template

Key

- Solid black = cut lines
- Dashed gray = fold / score lines
- Green brackets = measurements only

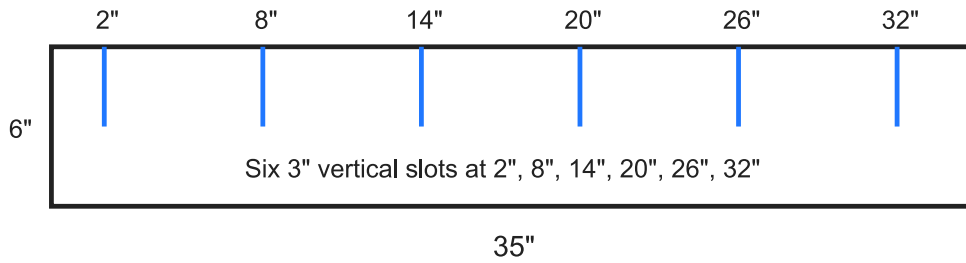


Pro Tip: This design is strongest when this panel is cut from one sheet (scored). But if that isn't available, you can cut these from smaller pieces (labeled A-E here and duct tape together)

Interior Side Supports and Center Spines

Side Support - Center Area

Qty 2

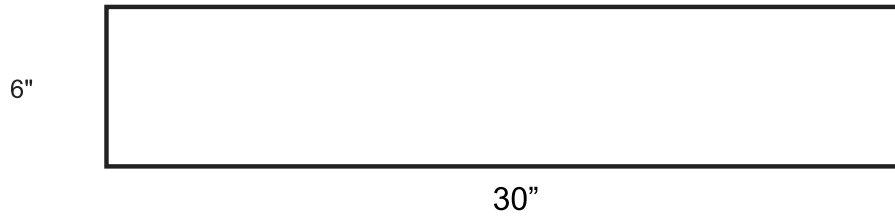


Key

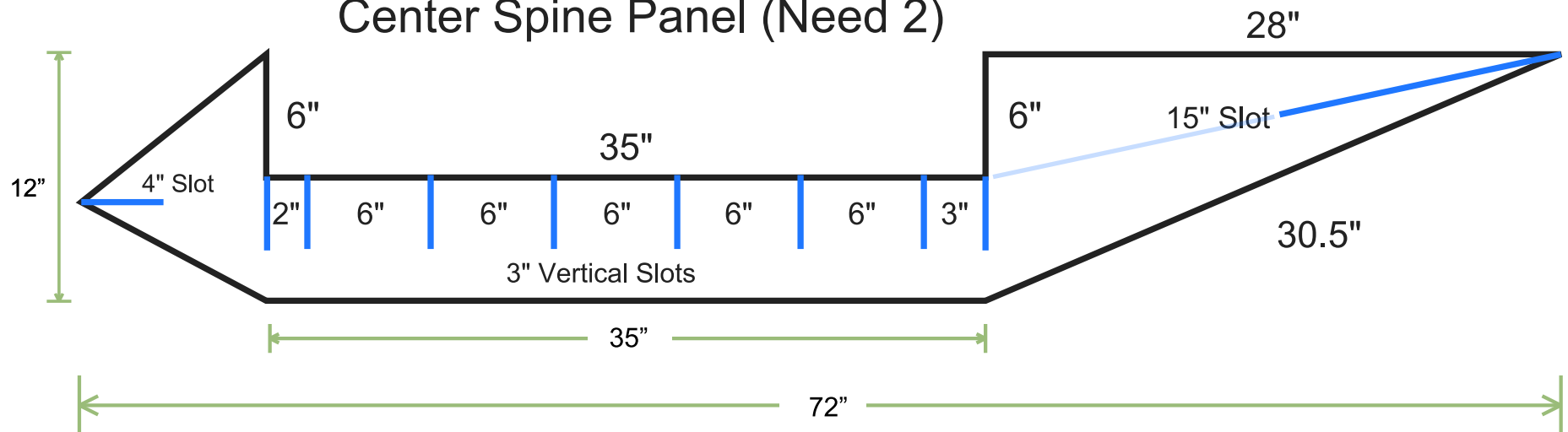
- Solid black = cut lines
- Measurements = Numbers only
- Blue line = slot cut

Cockpit Side Wall

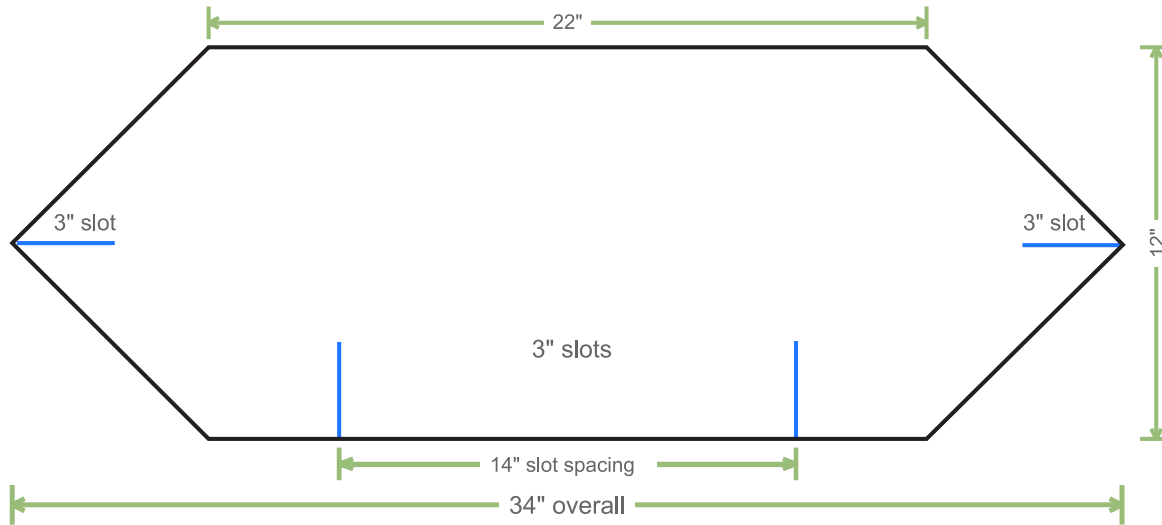
Qty 2



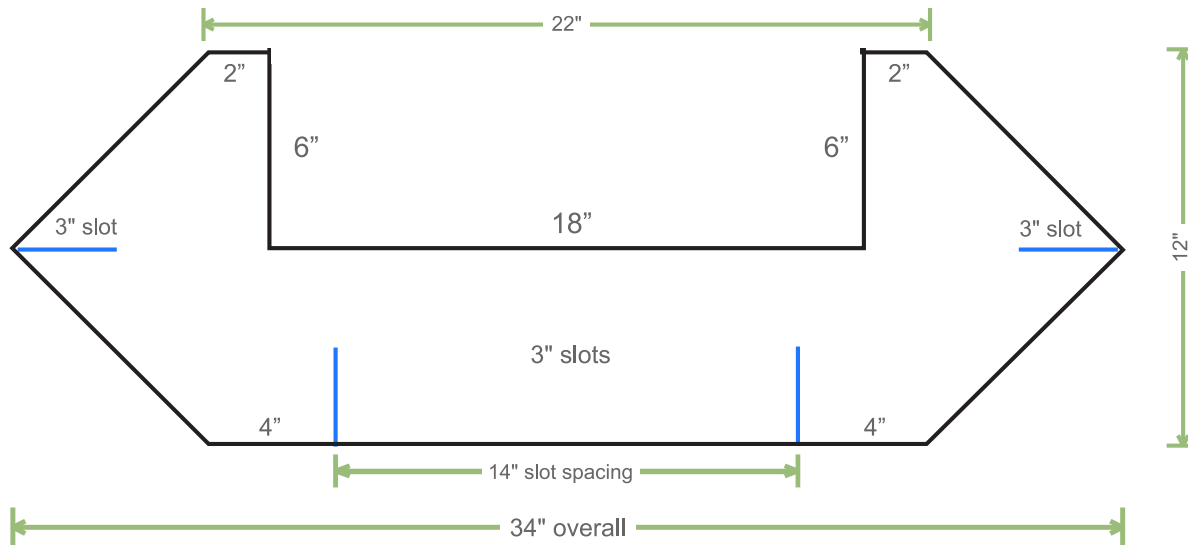
Center Spine Panel (Need 2)



Front & Back Internal Skeleton Panel A Need 4 pcs



Internal Skeleton Panel B Need 4 pcs



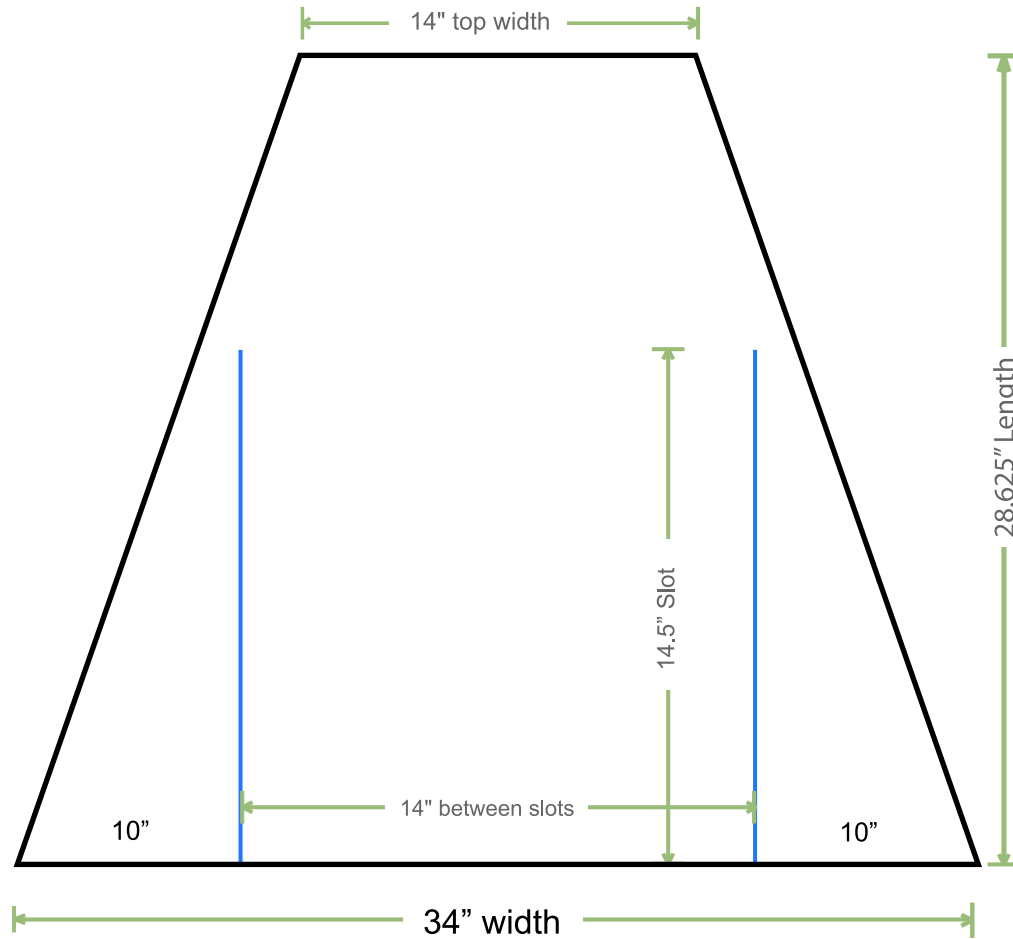
Key

- Solid black = cut lines
- Dashed gray = fold / score lines
- Green brackets = measurements only
- Blue line = slot cut

Top-view center sequence

A		2"
A		6"
B		6"
B		6"
B		6"
B		6"
A		6"
A		3"

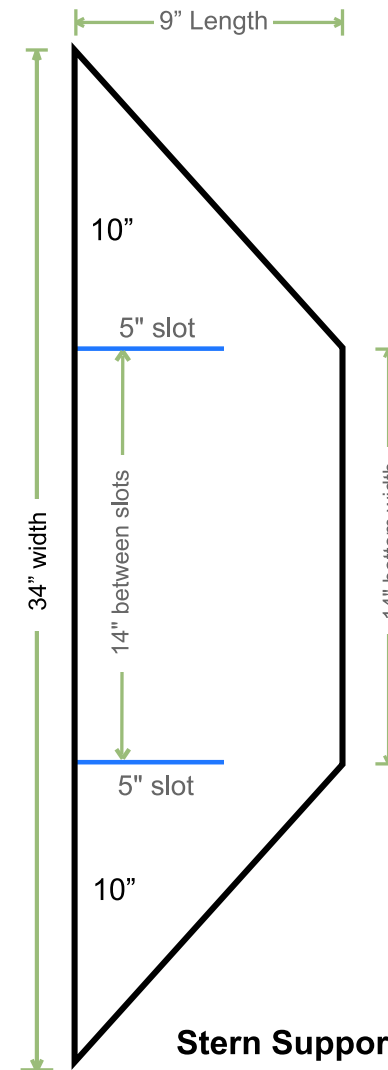
14" slot zone



Bow Support (Middle) Panel (x1)

Key

- Solid black = cut lines
- Dashed gray = fold / score lines
- Green brackets = measurements only
- Blue line = slot cut



Stern Support Panel (Middle) (x1)

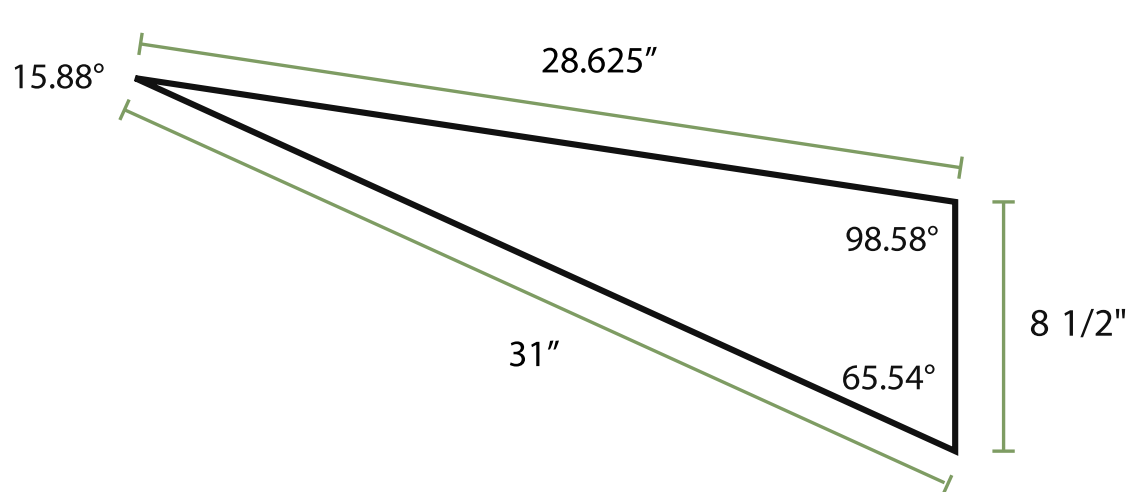
Bow & Stern Side Panels

Key

- Solid black = cut lines
- Dashed gray = fold / score lines
- Green brackets = measurements only

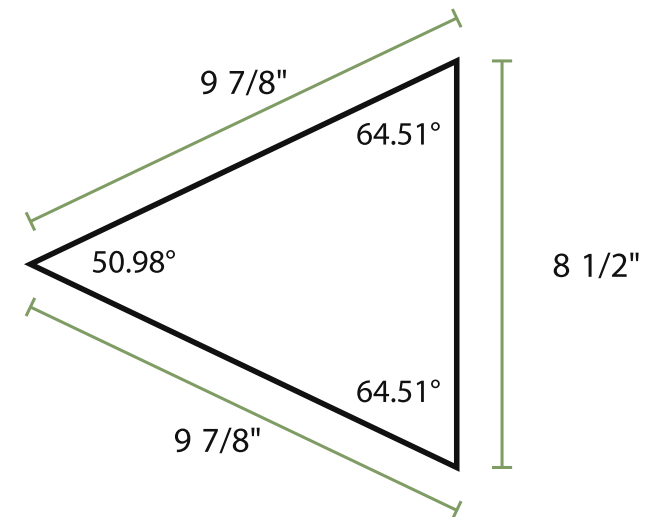
Bow side panel

Qty 2 mirrored



Stern side panel

Qty 2 mirrored



Pro Tip: You don't have to measure these... You can build the rest of the boat first, then overlay cardboard rectangles over each area to sketch the edges and then cut them to shape. (much easier this way) They should match so you only have to trace two and then use the cut ones to make a second mirrored panel. (affix with duct tape)